Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Product engineer and AI builder with 8+ years of end-to-end ownership -- from scoping and building to shipping, measuring, and iterating on systems used by 100K+ external users. I build 0-to-1 AI products: AgentProbe is a production agent harness and eval toolkit (5-metric LLM-as-judge, CI/CD regression gate that automatically blocks failing runs, golden datasets); Leasel is an 8-agent LangGraph pipeline with HITL escalation, reliability tracking, and guardrails built for knowledge workers (support engineers) across 1,240+ compliance test cases. Strong full-stack across FastAPI, React, PostgreSQL, and Python. Hands-on with LLM APIs (OpenAI, Anthropic, Groq), agent frameworks (LangChain, LangGraph), prompt engineering, tool integrations, and guardrail design. M.S. Data Analytics Engineering, GMU (GPA 3.87). AgentProbe: github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

0-to-1 Product Engineering | Eval Harnesses & Quality Gates | Agent Loops & Guardrails | HITL Design & Agent Reliability | LLM APIs & Agent Frameworks | Full-Stack (FastAPI, React, PostgreSQL) | Prompt Engineering | SQL & Product Metrics

WORK EXPERIENCE

Blue Horse Digital

Sep 2026 -- Present

AI/ML Engineer (Volunteer) -- Virginia, USA

- Building ML models and AI pipelines integrating data systems with back-end APIs and reporting layers; iterating based on application behavior and team feedback.

Magizhs Tech Solution

May 2025 -- Aug 2025

AI/ML Engineer Intern

- **Designed and shipped an end-to-end agentic automation pipeline** (LangGraph stateful graph: parse -> plan -> retrieve -> validate -> conditional routing) -- translating fuzzy survey requests into structured flight plans with grounded, risk-aware compliance decisions automatically.
- Implemented guardrails enforcing FAA altitude limits, battery budgets, and no-fly-zone checks -- blocking non-compliant outputs before they reached users, with clear failure messaging when blocked.
- Built a RAG layer (FAA Part 107 + operational corpus) with cited source retrieval -- every compliance check grounded in source regulation, not hallucinated.

George Mason University

Aug 2024 -- May 2026

Graduate Teaching Assistant + Research Associate

- Built multi-source prediction pipelines (Python, R) on WHO + UK Road Safety datasets (R2 = 0.977); applied Isolation Forest for automated data integrity checks; designed Tableau/Quarto dashboards tracking research metrics (+40% faculty comprehension).
- Fine-tuned a custom classification model (HuggingFace Transformers + PyTorch); deployed inference API on Quart; instrumented outputs and presented evaluation results to faculty stakeholders.

Entrata (PropTech / Real Estate SaaS)

Jan 2017 -- Aug 2024

Senior Software Engineer -- Lehi, UT (Remote) [promoted from Associate, 2018]

- **Shipped 20+ production modules to 100K+ external users across 120+ Agile sprints** -- end-to-end ownership: scope, build, ship, measure (SQL analytics on eligibility and financial workflows), and iterate based on production feedback.
- **Built and owned the RabbitMQ-based async processing pipeline** -- designed retry logic and idempotency guarantees; diagnosed and resolved reliability issues (double-processing failures, race conditions); zero compliance violations across 7 years of production release cycles.
- **Instrumented 200+ SQL queries to surface data correctness and performance metrics** -- 30% performance gain; caught 2 eligibility correctness bugs during analysis that unit tests missed; used production metrics to drive prioritization.
- Led AWS S3 migration for Affordable Housing module -- security controls (Libsodium-encrypted pipelines), 100% PHI compliance, zero incidents; eliminated 80% of manual operational workload via automation pipelines.
- Mentored 5+ engineers; conducted 300+ code reviews; built internal debugging tools that cut team debugging time by 40%.

PROJECTS

AgentProbe: AI Agent Evaluation Harness

LangGraph -- LangChain -- FastAPI -- ChromaDB -- React

Built a production-grade AI agent harness and eval toolkit from 0-to-1: 15-endpoint FastAPI backend, 5-metric LLM-as-judge quality gate (correctness, groundedness, latency, token efficiency, hallucination rate), CI/CD regression harness with golden datasets that automatically blocks failing agent runs before production, multi-provider LLM Router (OpenAI/Anthropic/Groq/Ollama), React observability dashboard with per-run regression drift detection. Measures task success rate per run. Used to evaluate Leasel across 1,240+ compliance test cases. 100% local, zero vendor cost.

github.com/venkatalakshmiparimala/AgentProbe

Leasel: Agentic Compliance Investigation Platform LangGraph 8-agent -- HITL -- FastAPI -- TF-IDF RAG -- Streamlit

0-to-1 agent harness for knowledge workers (support engineers): **Support Investigation** -- 8-agent LangGraph StateGraph (Triage -> Retrieval -> Config -> History -> Reasoning -> Validation -> HITL Escalation) producing ranked root-cause hypotheses; reliability and task success rate tracked via confidence scoring and eval harness (AgentProbe); **Document Intelligence** -- OCR-fallback ingestion, TF-IDF retrieval, citation grounding, human-review gate; **Tenant Self-Service** -- guardrail: eligibility questions always escalate to staff. Core design rule: LLM reasons; deterministic logic makes every final compliance call. Results: 93.6% precision / 87.9% F1 on 1,240 lease PDFs; 100% citation correctness.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AI on FHIR: NLP Clinical Records Query System Flask -- LLMs -- BERT -- React -- Docker -- FHIR

Designed and built a clean, usable product interface (Flask + React) without a dedicated designer -- NLP platform translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings; reduced manual clinical retrieval time ~60%; HIPAA-compliant RBAC; iterated based on clinical staff feedback.

github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering -- George Mason University — May 2026

GPA 3.87 -- Virginia, USA

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2017

SKILLS

AI & Agents: LangChain, LangGraph, Agentic AI, Agent Loops, Tool Integrations, Guardrails, HITL, LLM Evaluation, Eval Harnesses, LLM-as-Judge, Prompt Engineering, RAG Pipelines, ChromaDB, Embeddings, LLM Routing, OpenAI API, Anthropic API, Groq, Ollama, FastAPI, Agent Reliability, Observability, Hallucination Detection, MCP

Full-Stack: Python, JavaScript, TypeScript, React.js, Node.js, PHP, Java, SQL, PL/SQL, R, Django, Spring Boot, Flask, Streamlit, HTML/CSS

Cloud & Infra: AWS (S3, EC2, Lambda, Glue, Bedrock), Docker, Kubernetes, GitHub Actions CI/CD, PostgreSQL, RabbitMQ, Redis, Elasticsearch

ML & Data: PyTorch, HuggingFace Transformers, scikit-learn, XGBoost, Random Forest, Isolation Forest, Pandas, NumPy, Tableau, Jupyter